

Optimality conditions: unconstrained problems

Quadratic functions

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Optimization: principles and algorithms



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Quadratic functions

*Quadratic functions are such that the second derivative matrix is constant.
We investigate here the implications of the optimality conditions in this specific case.*

Quadratic function

Formulation

$$\min_{x \in \mathbb{R}^n} f(x) = \frac{1}{2} x^T Q x + g^T x + c$$

$$Q \in \mathbb{R}^{n \times n} \text{ symmetric}, g \in \mathbb{R}^n, c \in \mathbb{R}.$$

$$\nabla f(x) = Qx + g, \nabla^2 f(x) = Q.$$

Q is constant.

Necessary optimality conditions

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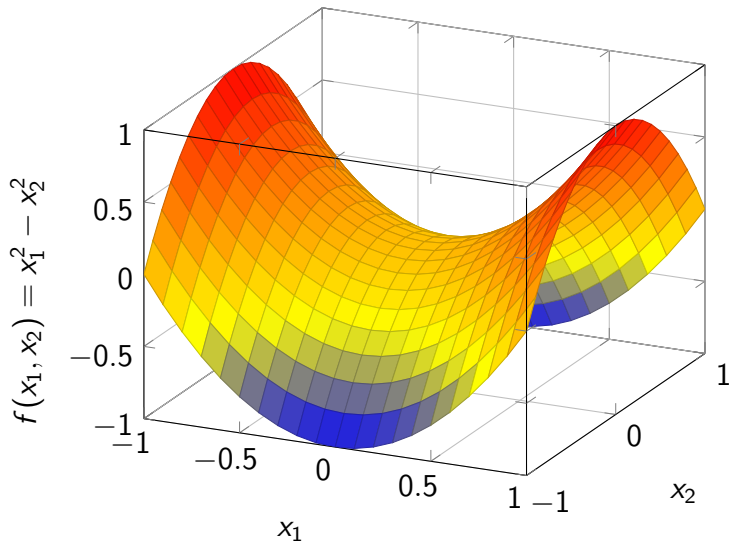
Necessary optimality conditions

- ▶ If Q is **not** positive semidefinite.
- ▶ At least one eigenvalue is negative.
- ▶ There exists a direction that is concave.
- ▶ The problem is unbounded.
- ▶ There is no optimum.

Necessary optimality conditions

Mention the eigenvalues: 1 and -1

Necessary optimality conditions



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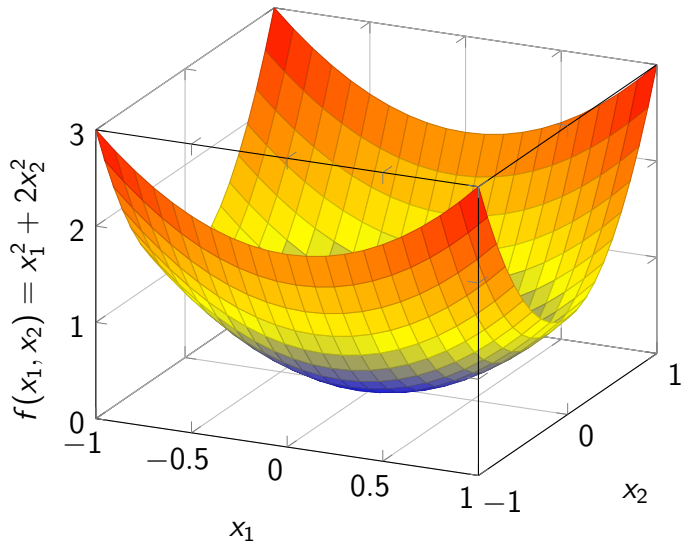
- ▶ If Q is positive definite,
- ▶ all eigenvalues are positive,
- ▶ the problem is strictly convex.
- ▶ The unique global optimum is obtained by solving

$$\nabla f(x^*) = 0 \iff Qx^* + g = 0.$$

Sufficient optimality conditions

Mention the eigenvalues: 1 and 2

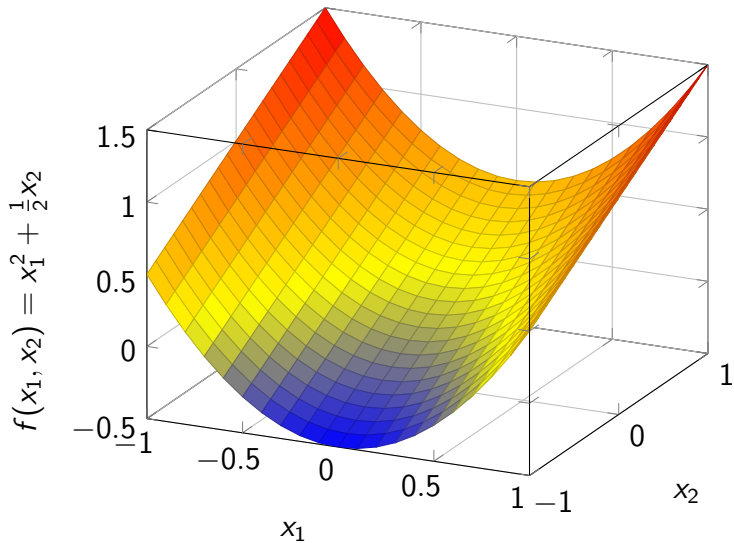
Sufficient optimality conditions



Last case

Positive semidefinite, but not positive definite. Eigenvalues: 1 and 0

Last case

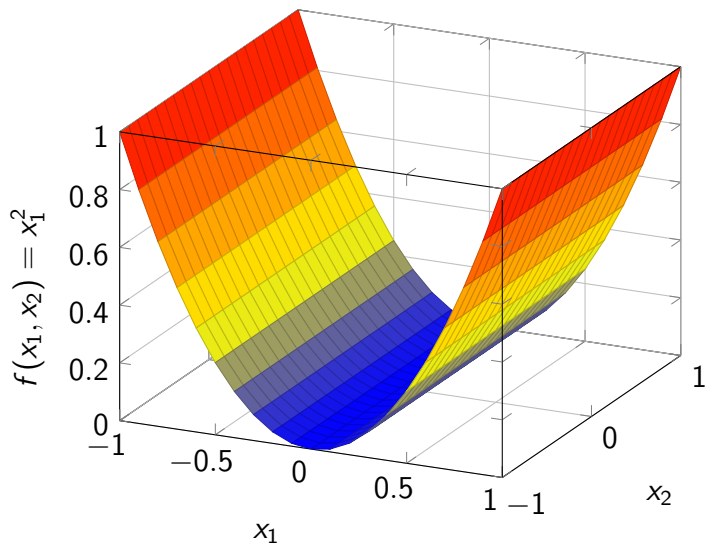


Last case

*Null eigenvalue means that the function has no curvature in this direction.
It is linear.*

*In unconstrained optimization, a linear function makes sense only if it is flat.
If not it is unbounded.*

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 - ▶ the function is unbounded, or
 - ▶ there is an infinite number of global maxima.

Summary

*Optimality conditions have been used to characterize a quadratic function.
The sign of the eigenvalues of Q characterize the curvature of the function
Positive: convex direction
Negative: concave direction, unbounded.
Zero: linear direction: flat or unbounded.*